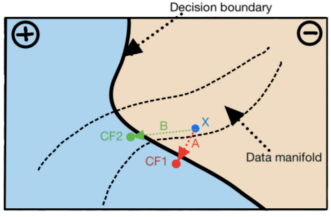
Interpretable Machine Learning

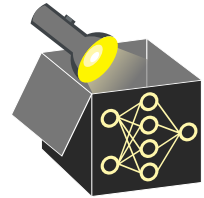
Counterfactual Explanations (CEs): Motivation



Learning goals

- Understand the motivation behind CEs
- Know why and how CEs are used
- Recognize the philosophical foundations of counterfactual reasoning

MOTIVATING EXAMPLE: CREDIT RISK & CE



x: customer and credit information

y: grant or reject credit

Age 52

Gender m

Job unskilled

Amount 10T

Duration 24

Purpose TV

→ **BLACK BOX** →

Grant

Reject

Potential questions:

- Why was the credit rejected?
- Is this decision fair compared with similar applicants?
- **How should x be changed so that the credit is accepted?**

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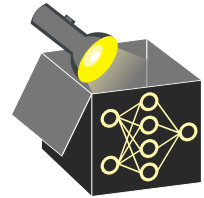
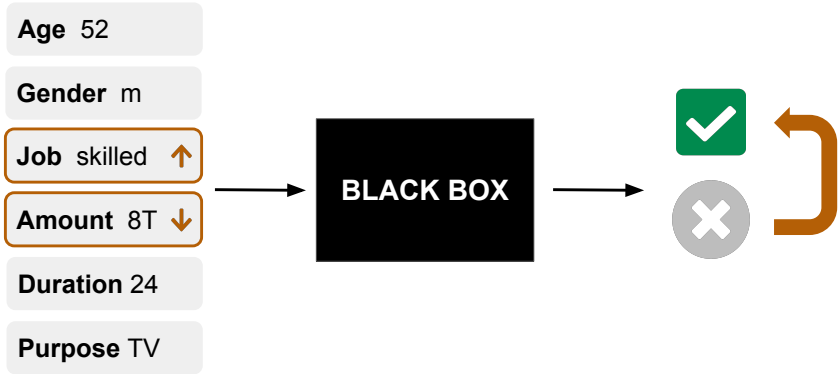
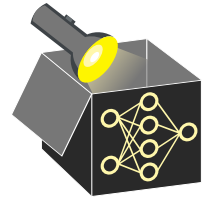


Counterfactual Explanations provide answers in the form of "What-If"-scenarios.



“If the applicant had higher skills and the credit amount had been reduced to $8.000, the loan would have been granted."

CORE DEFINITION AND PURPOSE OF CE



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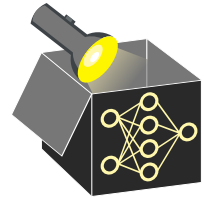

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$$\text{Find} \quad \mathbf{x}' \approx \mathbf{x} \quad \text{such that} \quad f(\mathbf{x}') = y' \quad \text{and distance} \quad d(\mathbf{x}, \mathbf{x}') \quad \text{is minimal}$$

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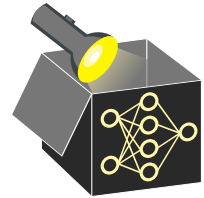

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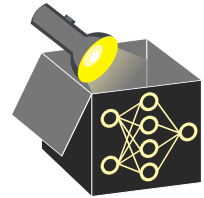

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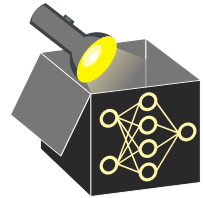
- **Minimal actionable changes:** Difference $\mathbf{x}' - \mathbf{x}$ shows the smallest feature change a user could realize in practice
- **Primary audience:**
  - Individuals aiming to alter model predictions
  - ML engineers exploring model behavior under adversarial conditions
    $\rightsquigarrow$ how small text changes in email flip prediction from "spam" to "no spam"

INTERPRETIVE AIMS & ROLES



CEs can serve various purposes; the user can decide what to learn from them, e.g.:

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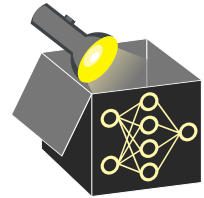

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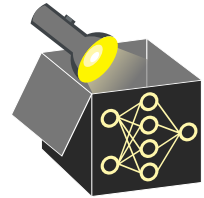

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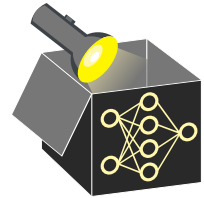

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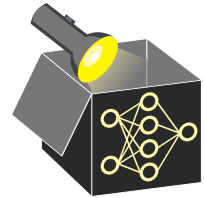

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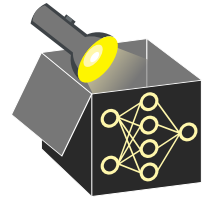
- **Detect model biases:**
  There is a bug, an increase in amount should not increase approval rates.

PHILOSOPHICAL FOUNDATIONS ▸ Lewis (1973)



Counterfactuals have a long tradition in analytic philosophy
$\rightsquigarrow$ A **counterfactual conditional** takes the form:

"If S had occurred, Q would have occurred."

- S: past event that never happened $\rightsquigarrow$ CE run contrary to fact
- Statement is true iff Q holds in all **closest** worlds where S is true
- Closest worlds preserve laws and change as few facts as possible (related to S)

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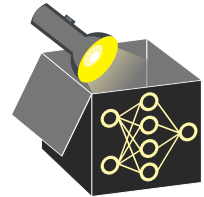

- CEs have largely been studied to explain causal dependence
- **Causal dependence**: Q depends on $S \Leftrightarrow$ without S, no Q
  $\rightsquigarrow$ Good CEs point to critical causal factors that drove the algorithmic decision
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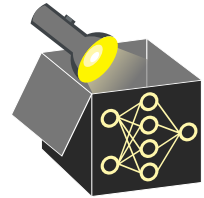

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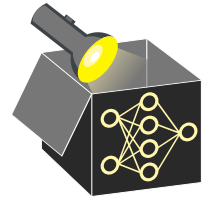

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- CEs are contrastive: Explain a decision by comparing it to a different outcome
  $\rightsquigarrow$ If age were 30 instead of 60, loan would have been $9k instead of rejected
  $\rightsquigarrow$ Answers contrastive question: “Why Q’ instead of Q?” (preferred by humans)